Michael de Oliveira

Quantum Computing Research Scientist | Foxconn

michaeldeoliveira848@gmail.com | www.linkedin.com/in/michael-de-oliveira | Amsterdam, Netherlands



PROFILE

Quantum computing research scientist with experience across industry and academia at Foxconn, IBM, Sorbonne University, and INL. Expertise in quantum algorithms, rigorous quantum advantage under realistic physical-resource constraints, circuit synthesis and complexity, fault-tolerant resource analysis, and formal quantum-hardware benchmarking.

RESEARCH AND INDUSTRY EXPERIENCE

Hon Hai (Foxconn) Research Institute **Mar 2026 - Present**

Research Scientist, Quantum Computing Research Center Amsterdam, Netherlands

- Conduct research on quantum-circuit resource complexity, including how circuit depth, gate set, and ancillary resources shape quantum computational advantage.
  - Co-authored work certifying quantum computational depth independently of gate set, circuit size, and ancillary qubits, and establishing provable advantages over shallow classical computation; accepted at FOCS 2026.
- Serve as project manager for a joint Foxconn-Oxford research programme, supervising three interns on quantum advantage, gate-set-dependent circuit scaling, and quantum-circuit simulation.

IBM **May 2025 - Aug 2025**

Quantum Computing Theory Research Scientist Intern Cambridge, Massachusetts, USA

- Co-developed quantum algorithms for PAC and agnostic learning, including a quantum agnostic boosting framework; published at COLT 2026 and presented at QIP 2026.

Hon Hai (Foxconn) Research Institute **Sep 2024 - Apr 2025**

Part-time Quantum Computing Researcher Taipei, Taiwan / Remote

- Authored a study proving unconditional quantum advantage for constant-depth circuits over shallow classical circuit models relevant to neural networks. Developed a surface-code-based fault-tolerant construction and analyzed resource requirements for physical implementation; published in Nature Communications.

Sorbonne University (LIP6) **Oct 2022 - Aug 2024**

Visiting PhD Student Paris, France

- Co-developed a hardware-agnostic, heuristic-free, verification-inspired benchmark for quantum circuits solving general decision problems, subsequently demonstrated experimentally on Quantinuum's H1 processor.
- Co-supervised the course 'Advanced Quantum Information Processing' in the MSc Quantum Information programme.

International Iberian Nanotechnology Laboratory (INL) **Sep 2021 - Dec 2025**

PhD Researcher in Computer Science Braga, Portugal

- Designed circuit-synthesis methods for non-adaptive measurement-based quantum computation, achieving exponential resource reductions over prior constructions with detailed quantum-classical gate-count comparisons.
- First-authored work establishing a quantum advantage in gate complexity under hardware-motivated assumptions on the available multi-qubit gate set.

EDUCATION

University of Porto / INL - PhD in Computer Science, 2025. Thesis: 'The Interplay between Quantum Foundations and Circuit Complexity'; defended 3 Dec 2025. Supervisors: Luís Soares Barbosa, Elham Kashefi, and Ernesto Galvão. Examination committee included Dan Browne (UCL) and Daniel Grier (UC San Diego).

- Thesis awarded the highest possible grade.

University of Minho - MSc and BSc in Engineering Physics.

- MSc final grade: ECTS A.
- Excellence Awards in two BSc years and the first MSc year.

RESEARCH SUPERVISION AND SERVICE

Research supervision: three current interns across Foxconn and the University of Oxford working on quantum advantage, gate-set-dependent circuit scaling, and quantum-circuit simulation.

MSc supervision: 'Quantum Bayesian Reinforcement Learning', awarded 20/20 (University of Minho, 2022).

Peer review: SODA 2027, Nature Communications, STOC 2026, QIP 2024, IEEE QSW 2024, YQIS 2024, and CCC 2023.

Research visits: Visited the research groups of Professors Robert König at the Technical University of Munich (2023), Hans Briegel at the University of Innsbruck (2023), and Jens Eisert at Freie Universität Berlin (2026).

TECHNICAL SKILLS

Programming and tools: Python, Qiskit, NumPy, Matplotlib, Wolfram Mathematica, C, Haskell, and Erlang.

SELECTED TECHNICAL PRESENTATIONS

- **Invited speaker:** 'A rigid fingerprint of quantum magic', Atom Computing, USA.
- **Invited speaker:** 'On the computational power of commuting and non-commuting operations for quantum processes', VeriQloud, France.
- **Invited speaker:** 'Temporally unstructured measurement-based quantum computation with advantage', Quantum Software Lab Workshop, Edinburgh.

RECOGNITION

New Talents in Quantum Technologies - Gulbenkian Foundation; one of eight national recipients in the award year.

SELECTED RESEARCH CONTRIBUTIONS

Worst-case depth hierarchy for shallow quantum circuits. **FOCS 2026 (accepted)**
Min-Hsiu Hsieh, **Michael de Oliveira**, Sathyawageeswar Subramanian, Xingjian Zhang

Unconditional advantage of noisy qudit quantum circuits over biased threshold circuits in constant depth. **Nature Communications 16, 3559 (2025)**
Michael de Oliveira, Sathyawageeswar Subramanian, Leandro Mendes, Min-Hsiu Hsieh

Learning depth-3 circuits via quantum agnostic boosting. **COLT 2026, PMLR 336:371-426; presented at QIP 2026**
Srinivasan Arunachalam, Arkopal Dutt, Alexandru Gheorghiu, **Michael de Oliveira**

Quantum Bayesian networks can speed up reinforcement learning in partially observable environments. **Quantum Machine Intelligence 8, 65 (2026)**
Gilberto Cunha, Alexandra Ramôa, André Sequeira, **Michael de Oliveira**, Luís Soares Barbosa

Heuristic-free verification-inspired quantum benchmarking. **Quantum Science and Technology 10(2), 025055 (2025)**
Johannes Frank, Elham Kashefi, Dominik Leichtle, **Michael de Oliveira**

Quantum advantage in temporally flat measurement-based quantum computation. **Quantum 8, 1312 (2024)**
Michael de Oliveira, Luís Soares Barbosa, Ernesto Galvão

The power of shallow-depth Toffoli and qudit quantum circuits. **ACM Transactions on Quantum Computing (2026)**
Alex Bredariol Grilo, Elham Kashefi, Damian Markham, **Michael de Oliveira**